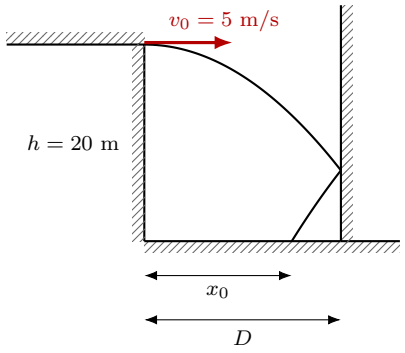
Kinematics 1D and 2D Revision Sheet

Level 4 Sheet **Maximum Time Taken to Solve: 40 Min**

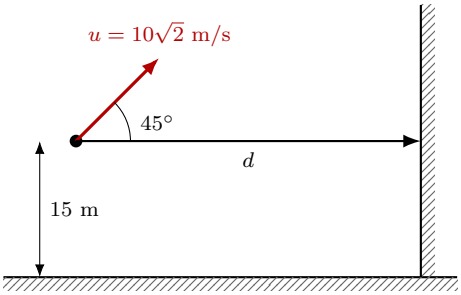
Note: Revise Class Notes + Do Last 5 Years of PYQ + These Sheets of Level 1 to 4.

1. A particle is projected from horizontal surface of earth at speed u making an angle $\theta = \cos^{-1}\left(\sqrt{\frac{5}{72}}\right)$ with the horizontal. If equation of trajectory for the particle is $y = kx - 2x^2$ (where k is a constant) then what is the speed (in m/s) of projection. (Assume that particle was projected from origin and motion of particle is in xy plane, where x-axis is along the horizontal direction and $+y$-axis is vertically upward) [Take $g = 10$ m/s^2]

2. A ball leaves a horizontal table with velocity $v_0 = 5$ m/s. The ball bounces elastically from a vertical wall at a horizontal distance D ($= 8$ m) from the table, as shown in figure. The ball then strikes the floor a distance x_0 from the table ($g = 10$ m/s^2). The value of x_0 (in metre) is



3. A small ball is thrown from a height of 15 m above ground and at a horizontal distance d from a vertical wall. The ball first hits the wall and then strikes the ground and then it flies back to its initial position of throwing. Take both collisions to be perfectly elastic and neglect friction. The initial speed of the ball is $10\sqrt{2}$ m/s and angle of projection is 45° with the horizontal as shown. Find the horizontal distance of point of throwing from the wall 'd' in meters. (Neglect air resistance and take $g = 10$ m/s^2)



4. A river of width 100 m is flowing with a velocity of 1.5 m/s. A man start from one end with rest relative the river. He rows with an acceleration of 2 m/s^2 relative to the river. If the man want to cross the river in minimum time, by how much distance (in meters) will he be drifted (flown) in the direction of river flow during the crossing.

5. A projectile is fired from horizontal ground with speed v and projection angle θ. When the acceleration due to gravity is g, the range of the projectile is d. If at the highest point in its trajectory, the projectile enters a different region where the effective acceleration due to gravity is $g' = \dfrac{g}{0.81}$, then the new range is $d' = nd$. The value of n is _____.

Keep Going!

"Every great projectile starts with the right angle and a strong push. Aim high, trust your preparation, and keep going."